

Information Visualization and Analysis (Autumn 2025-26)

1. Course Title	Information Visualization and Analysis
2. Credit Structure	Lecture hours per week: 3 Tutorial hours per week: 0 Practical hours per week: 2 Total Credits: 4
3. Course Code	IT405/ IT569
4. Program/Semester	B.Tech Sem VII and M.Tech Sem III
5. Category	Technical Elective
6. Prerequisite courses	Basic understanding of data and statistics
7. Foundation for	Data Science and analytics. Advanced Scientific and Engineering Visualization, Machine Learning interpretability
8. Abstract Content	This course provides a comprehensive introduction to the principles and practices of information and data visualization, combining theoretical concepts with practical skills. Students will explore key topics such as transforming complex datasets into effective visual representations that support analysis and critical thinking, human perception, fundamentals of visual encoding, color theory, decision-making, and storytelling through data. Multiple case studies across domains will illustrate the real-world impact of effective information visualizations. Hands-on lab assignments using tools like Tableau and realworld datasets will reinforce lecture concepts. A final project allow students to apply visualization techniques to complex datasets and developing data-driven storytelling skills.
9. Course Materials/ References Suggested Text book(s)	• Information Visualization: Perception for Design by Colin Ware. Morgan Kaufmann • Better Data Visualizations - A Guide for Scholars, Researchers, and Wonks, by Jonathan Schwabish. Columbia University Press • Visualization Analysis and Design by Tamara Munzner. CRC Press. • The Visual Display of Quantitative Information by Edward R. Tufte .

10. Evaluation/ Grading Policy	Two Exams (55 marks): One Mid sem exam and Final end-sem examination. (20+35 marks) Lab Assignments: There will be around 6-7 Lab assignments in the whole course comprising a total 30 marks. Lab attendance is compulsory. Lab attendance will carry 20% marks in each assignment. 1 course Project: 15 marks (Project may be evaluated in multiple ways). Grading scheme is relative and depends on both: class performance and minimum expectation from a student.
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Detailed Course Contents		
Topic Name	Content (2 -3 lines per 4 – 6 lectures)	No. of lectures
Module 1	Introduction to Information and Data visualization. Human perception and cognition, Gestalt principles. Historical context, importance and key applications using case-studies such as Anscombe's quartet, John Snow's work on the cholera outbreaks, Charles Minard's chart etc. Getting started with Tableau	5
Module 2	Data Types, Structures and Semantics. Data Abstraction - Dataset Types (Tables, fields etc), Attribute Types (Categorical, Ordered etc), Nominal, ordinal, quantitative data, interval, multidimensional, hierarchical, temporal, spatial data. Data preparation and review of important EDA concepts including visualizing distributions.	5
Module 3	Task abstraction. Types of tasks. Actions and Targets, Analyzing and Deriving. Relating tasks to data. Interaction and visual encoding for task support. Seeing relationships.	4
Module 4	Marks and Channels - Using Marks and Channels, Visual Channels and encoding, Data Type to Channel mapping, Channel Effectiveness. Visual Encodings in practice, bad vs. good Encodings with case studies.	4
Module 5	Spatial organization, layout design, and visual encoding of data and values. Arrangements by Keys and Values, Express Quantitative values, Separate, Order, Align, Spatial Axis Orientation, Spatial Layout Density. Arrange Spatial fields - Geometry, Geographic Data, Scalar Fields, Vector Fields. Encoding Hierarchy and Grouping, Scales and Axis Design.	5

Module 6	Color scales and datatypes- Theory, HSL, colormaps, categorical vs ordered color, color models, color palettes, color deficiency, color contrast and discriminability.	3
Module 7	View Manipulation and visual interaction - Zoom, pan, filter, brushing, linking. Change View over Time, Select Elements, Navigate, facet into Multiple Views. Reduce Items and Attributes, hierarchical and multivariate data. Ethics and Bias in Visualization - Misleading visuals, data bias.	4
Module 8	Information visualization choices, design principles, presentation and storytelling techniques using multiple visualization idioms and case-studies - Scatterplot, bar and line chart, streamgraph, stacked bar chart, Gantt charts, slope graphs, heatmap, histogram, box-plot, violin plot, density plot, streamgraph, treemap, Sankey diagram, ridgeline plot, radar plot, pie and coxcomb chart, glyph maps, SPLOM, parallel coordinates, choropleth, symbol maps, cartograms, dot density maps, isosurfaces, tooltips, small multiples, MAUP, Treemaps, flow maps. dendrograms, Node-link diagrams.	6
Advanced Topics	Uncertainty Visualization, Interactive Storytelling, Visualization for Machine Learning. Case studies. <i>Based on availability of time and how the courses progresses, data visualization based thematic paper discussion are also included in the lecture sessions.</i>	4

- Course Outcome:

CO1: Understand and apply the principles of human perception, data types, and encoding techniques to create meaningful and accurate visual representations of data and information.

CO2: Analyze data-driven tasks and select appropriate visualization techniques and interaction methods to support exploration, comparison, and communication of insights across varied real-life datasets.

CO1: Critically evaluate, design, and present ethical, effective, and insightful visualizations using latest tools (e.g., Tableau) and case-study-driven storytelling techniques.

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	x		x	x				x	x		x